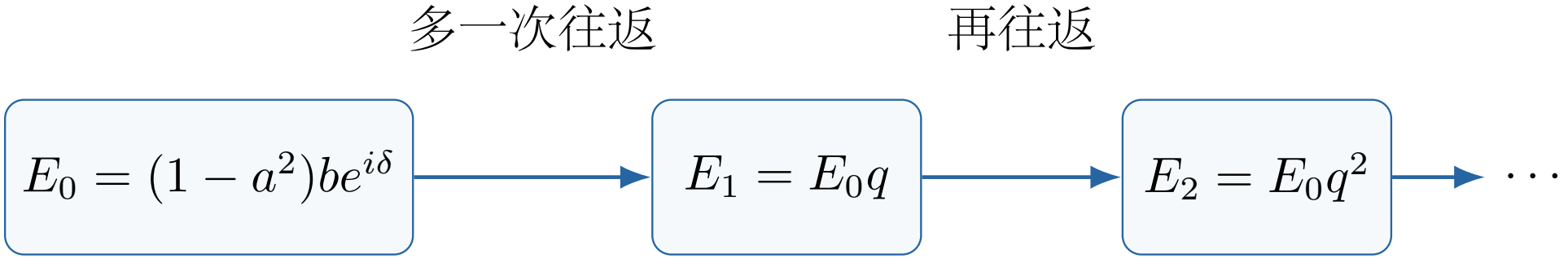

$$q = r_{10} r_{12} e^{i\delta} = -abe^{i\delta}$$